

# Algorithms as Recipes

Project: Helpful Classroom Sorter

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Features: Useful Clues

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Sound Becomes a Signal

Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Sensors Are Messengers

Without opening the old lesson, explain one key idea from Sensors Are Messengers. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Distinguish replacement from addition for Algorithms and changing state. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Algorithms and changing state. Proposed explanation: “Every instruction uses the original number”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Trace two changing number boxes

For a new display, X starts at 4 and Y starts at 1. Copy the current X into Y; add 3 to X; double Y; report both and stop. Record X and Y initially and after each instruction. Explain why Y does not become 7 when X changes. Then swap the first two instructions and trace the new procedure from the same initial state.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. Trace position and direction after every instruction. What if TURN were unspecified?

## 6. Solve the given case

A robot starts on the left of three squares, facing right. It executes MOVE, MOVE, STOP, each MOVE being one square.

Trace the position after each command.

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

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## 7. Check a changed case

Instructions say pick up a pencil, write your name, then find paper. Find the first ordering problem and rewrite the three steps so another learner can follow them.

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 9 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_

\_\_\_\_\_

## 9. Project checkpoint

Write the first executable sorting algorithm for one easy item before adding branches or AI.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

On numbered squares 0–5, start at 1 facing increasing numbers. Execute FORWARD 2, TURN, FORWARD 1, STOP. Define TURN as reversing direction. Trace position and direction after every instruction. What if TURN were unspecified?

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 Trace the current value

Start score at 3. Add 4 to score, double score, then subtract 5. Record the initial state and the state after each instruction. Underline the number used by the doubling step and explain where it came from.

### My reasoning and evidence

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## 13. W2 Distinguish replacement from addition

Two procedures start score at 6. Procedure A sets score to 4, then adds 2. Procedure B adds 4 to score, then adds 2. Trace both and explain why the first instructions are different.

### My reasoning and evidence

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## 14. W3 Repair the first error

A procedure starts score at 6, adds 2, doubles score, then subtracts 3. A trace says 6, 8, 12, 9. Identify the first wrong transition, explain the likely mistake and write a complete corrected trace. Why is changing only the last value inadequate?

### My reasoning and evidence

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## 15. W4 Trace two changing number boxes

For a new display, X starts at 4 and Y starts at 1. Copy the current X into Y; add 3 to X; double Y; report both and stop. Record X and Y initially and after each instruction. Explain why Y does not become 7 when X changes. Then swap the first two instructions and trace the new procedure from the same initial state.

### My reasoning and evidence

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## 16. W5 Test what a final answer can prove

The required procedure starts score at 10, adds 2 and subtracts 2. A learner instead adds 4 and subtracts 4. Both report 10. Write both traces and explain whether matching the final result proves that the learner followed the required procedure.

### My reasoning and evidence

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## 17. Feedback, return check and learning record

### Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_  
\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Algorithms as Recipes

A fictional paper badge machine transforms an initial score using an ordered recipe.

Start score=3. Recipe A: double, add 4, subtract 1, report.

Recipe B swaps the first two operations but keeps the same starting score.

## Changing instruction order

|          | Start | After step 1 | After step 2 | Report |
|----------|-------|--------------|--------------|--------|
| Recipe A | 3     | 6            | 10           | 9      |
| Recipe B | 3     | 7            | 14           | 13     |

Read a cell using both its row and column labels.

## Algorithms as Recipes

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Start

Both runs begin at 3.



#### 2. Recipe A

3 -> 6 after doubling -> 10 after adding 4 -> 9 after subtracting 1.



#### 3. Recipe B

3 -> 7 after adding 4 -> 14 after doubling -> 13 after subtracting 1.



#### 4. Compare outputs

Same operations, different order: outputs 9 and 13.

### CHANGE ONE THING

Change the starting score to 0 for both recipes.

Explain the changed result using the matching lesson.

Use the arrows to explain where the two zero-start runs first differ.

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Write the sorter's steps in order, showing which values each step updates so a partner can reproduce the result.

# 19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

Write the sorter's steps in order, showing which values each step updates so a partner can reproduce the result.

## Evidence to keep · My record

**Evidence to keep:** This week's input and deciding step

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** Prediction and actual result (or labelled paper trace)

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One change since the previous checkpoint and why

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence to keep:** One error or limit and the next test

**My record:** \_\_\_\_\_  
\_\_\_\_\_

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## 20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper badge machine transforms an initial score using an ordered recipe.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Same operations, different order: outputs 9 and 13.

1. Recipe A
2. Compare outputs
3. Recipe B
4. Start

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Compare outputs Same operations, different order: outputs 9 and 13. Every instruction reads the current score, not the original one unless explicitly told to. In B, doubling also doubles the added 4. In A, the extra 4 is added after doubling. The final difference of 4 follows from that order, not from different starting data.

### Practice 2: explain the deciding evidence

Changed-case practice: Change the starting score to 0 for both recipes. Which conclusion is supported by the supplied facts?

1. A reports 3 and B reports 7.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

#### My selected conclusion, deciding fact and explanation

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: A reports 3 and B reports 7. A:  $0 \rightarrow 0 \rightarrow 4 \rightarrow 3$ . B:  $0 \rightarrow 4 \rightarrow 8 \rightarrow 7$ .

## Repair a missing building block

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early input.. Return to this week and explain what you changed.

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered sequence of sound-signal samples with a stated time interval. Short practice: Does the all-zero retained record prove the full signal was silent? Use the picture.. Return to this week and explain what you changed.

Review Sensors Are Messengers (ai-1/3). Goal: Describe a measurement using its quantity, value, unit and conditions. Short practice: Why can we calculate the earlier gap but not predict an exact new B reading?. Return to this week and explain what you changed.

## Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

### What I repaired, the evidence I kept and the next test

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